

Nour Belgacem

Electrical Engineer

Power Electronics / Power Converters / Advanced Electrical Systems

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WORK EXPERIENCE

- **Developed Multi-Protocol Interfaces for a Test Bench in Qt/C++** *February 2026 – July 2026*
  Final-Year Internship SERMA International, Tunisia
  - Developed Qt/C++ interfaces over CAN/CAN FD and RS-232/UART for real-time RAM/FRAM data acquisition with sub-millisecond timing accuracy.
  - Built an aeronautical equipment qualification interface (NI USB-6001) with automatic test report generation.

  Key skills: CAN/CAN FD, RS-232/UART, Qt/C++, NI USB-6001, Embedded systems, Aeronautics, Scrumban, Jira.

- **Designed a Cable Cut-Detection Circuit Board** *April 2025 – May 2025*
  Year-End Project (PFA) ENIT, Tunisia
  - Designed a cable cut-detection board end to end (schematic, PCB routing, prototype) and developed the PIC12F675 firmware in C under MPLAB IDE.
  - Sized each stage (Schmitt trigger, voltage dividers, comparator, LC filter, Villard doubler), validated it in PSpice, then tested it on the fabricated board.

  Key skills: PIC12F675, Embedded C, MPLAB IDE, PICkit, PCB design, PSpice, Schmitt trigger, Voltage comparator, LC filter, Villard voltage doubler, Circuit testing.

ACADEMIC PROJECTS

- **Study and Modeling of a Bidirectional GaN Charger for Electric Vehicles** *September 2025 – January 2026*
  Academic Project ENIT, Tunisia
  - Studied and modeled a 4-quadrant DC/DC converter based on GaN components for bidirectional EV charging, driving an inductive (RL) load.
  - Designed a PI current regulator on the first-order RL model, achieving zero steady-state error, fast response, and validated four-quadrant operation in simulation.
  - Analyzed the 10 kHz PWM current ripple and prepared the real-time rapid-prototyping implementation for test-bench validation.

  Key skills: GaN, DC/DC converters, 4-quadrant converter, PI control, Current regulation, PWM.

- **Designed a Control Board for a Flyback Converter** *September 2025 – November 2025*
  Academic Project ENIT, Tunisia
  - Designed an isolated Flyback switch-mode power supply delivering regulated DC outputs (15V from a 90V input): defined the specifications, sized the transformer, rectification and filtering stages, built the SG3525 PWM control circuit, and routed the board on EasyEDA following EMC rules, then validated efficiency and dynamic stability in theory and in practice.

  Key skills: Flyback converter, SG3525, PWM control, Transformer design, EMC, EasyEDA, Power electronics.

EDUCATION

- **National Engineering School of Tunis (ENIT)** *September 2023 – June 2026*
  Engineering Degree in Electrical Engineering – *Advanced Electronic Systems (SEA)* Tunis, Tunisia

TECHNICAL SKILLS

- **Simulation tools:** LTspice, PSIM, Proteus, ModelSim, MATLAB/Simulink
- **PCB design tools:** EasyEDA, Eagle, KiCad

LANGUAGES

French: B2 (DELF) **English:** B2 (TOEIC: 785)